

Cerebrum: Serverless Model Deployment Step-by-Step Guide



Cerebrium is a cloud GPU infrastructure platform designed to simplify the deployment and scaling of machine learning applications. By managing complex infrastructure—GPUs, Kubernetes, queues, monitoring, and scaling—Cerebrium enables teams to focus on building impactful applications.

The platform offers rapid cold starts (under 5 seconds), a variety of GPU options, automated scaling from 1 to over 10,000 requests, and an intuitive deployment process. With 99.9% uptime and ongoing weekly enhancements driven by user feedback, Cerebrium ensures the reliability required for production-grade ML applications.

So let's get started 🚀 We would be deploying a small llama3.2 1B param model (Folder download link at the end)

1. Create a Cerebrium account

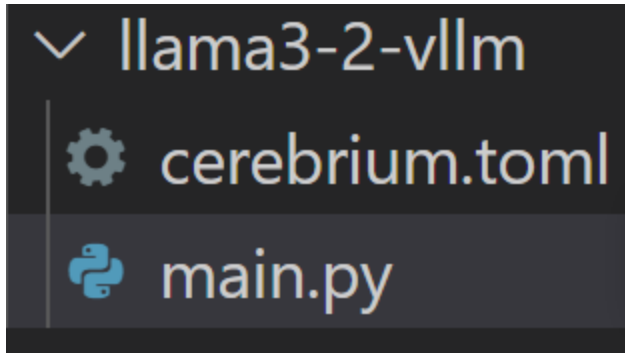
- Login to the cerebrium.ai
- Cerebrium provides \$30 dollars on a free trial

2. Install the Cerebrium library and CLI login

- Use pip install cerebrium
- `cerebrium login` # Redirects to your browser for authentication

3. Initialize the project

- `cerebrium init <project name>`
- Upon running the command a folder shall be automatically created like this



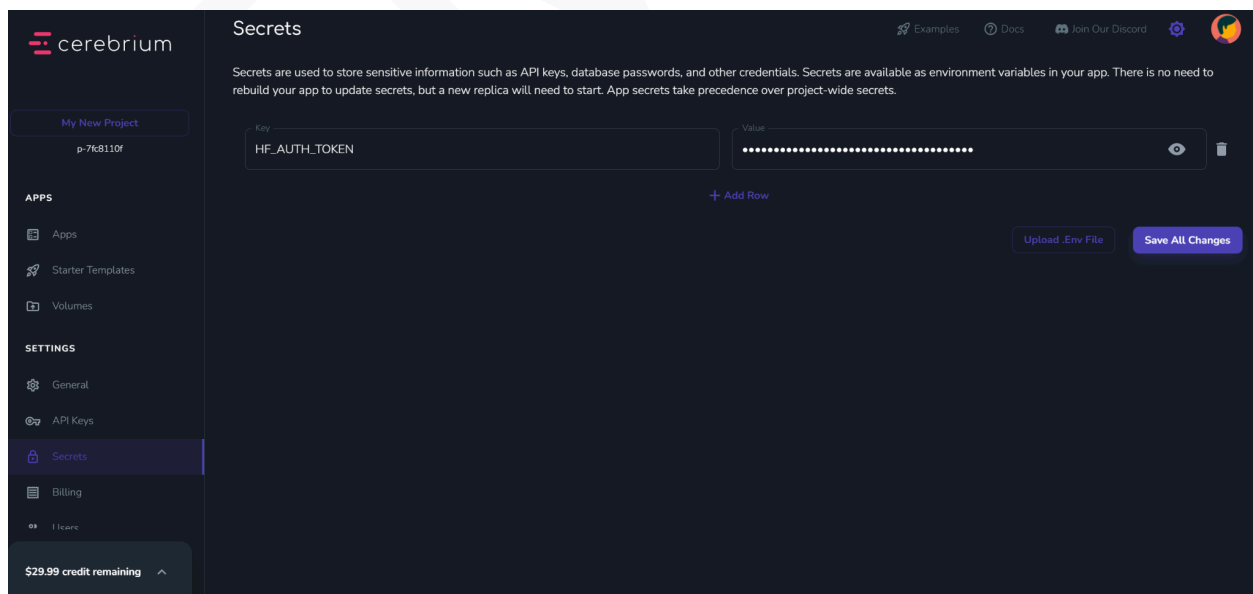
- Change the current working directory using cd.
cd <project folder>

4. File structure

- The main.py would include all the main code files be it a model, a rag system or an agentic system
- cerebrium.toml consists of the information regarding the app, and the hardware setup like GPU, vRAM, memory etc. It also includes the list of libraries that need to be installed.

5. Secrets

On the cerebrium dashboard add a secret 'HF_AUTH_TOKEN'

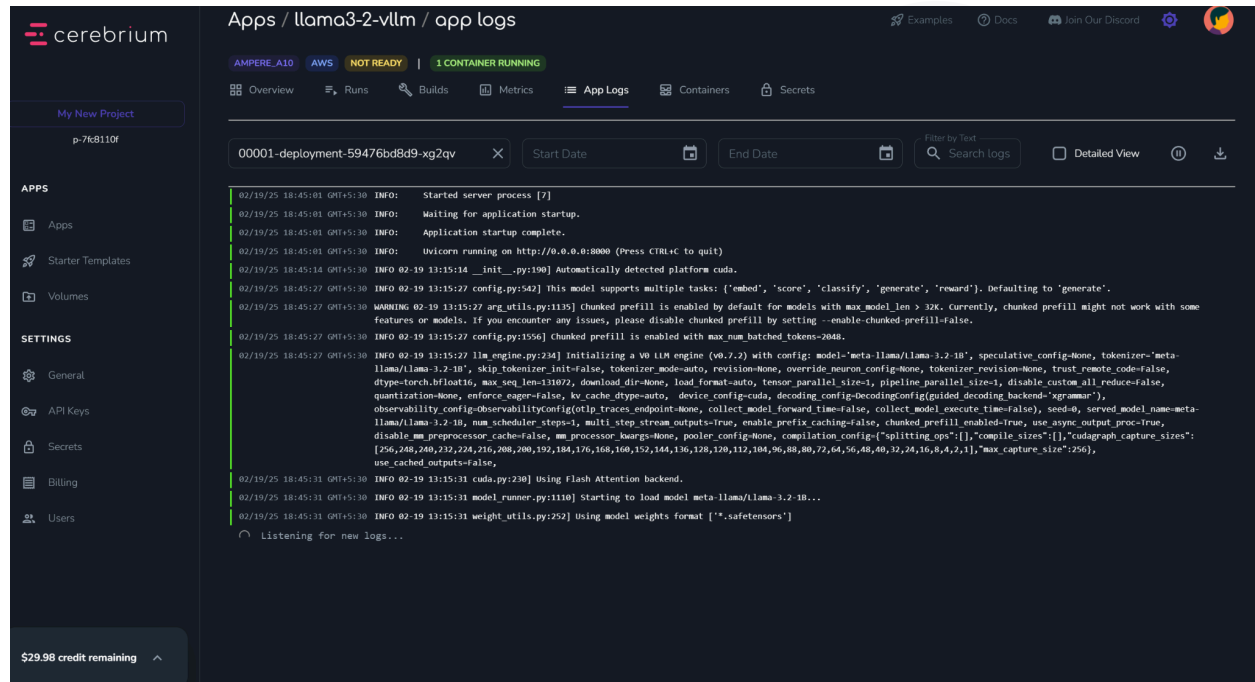


5. Deploy the app

Run cerebrium deploy

6. Building

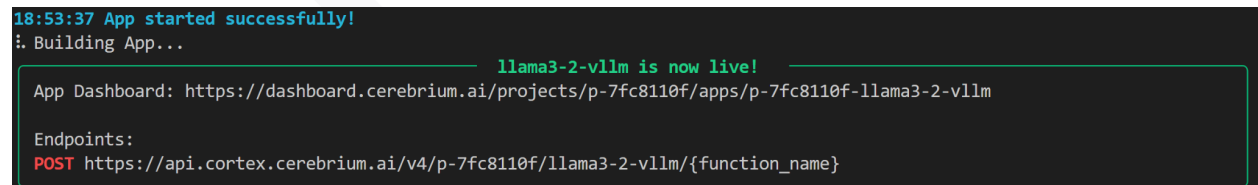
After running the above command one can check the app build status and check the logs



The screenshot shows the Cerebrium dashboard interface. On the left is a sidebar with navigation options: 'My New Project' (p-7fc8110f), 'APPS' (with sub-items: Apps, Starter Templates, Volumes), 'SETTINGS' (with sub-items: General, API Keys, Secrets, Billing, Users), and a '\$29.98 credit remaining' indicator. The main panel is titled 'Apps / llama3-2-vllm / app logs'. It features a status bar at the top indicating 'AMPERE_A10', 'AWS', 'NOT READY', and '1 CONTAINER RUNNING'. Below this is a tabbed interface with 'App Logs' selected. The logs display a series of messages from 02/19/25 18:45:01 to 02/19/25 18:45:31, including server startup, application completion, and detailed configuration for the llama3-2-vllm model. A search bar and filters are visible at the top of the log panel.

7. Deploy successful

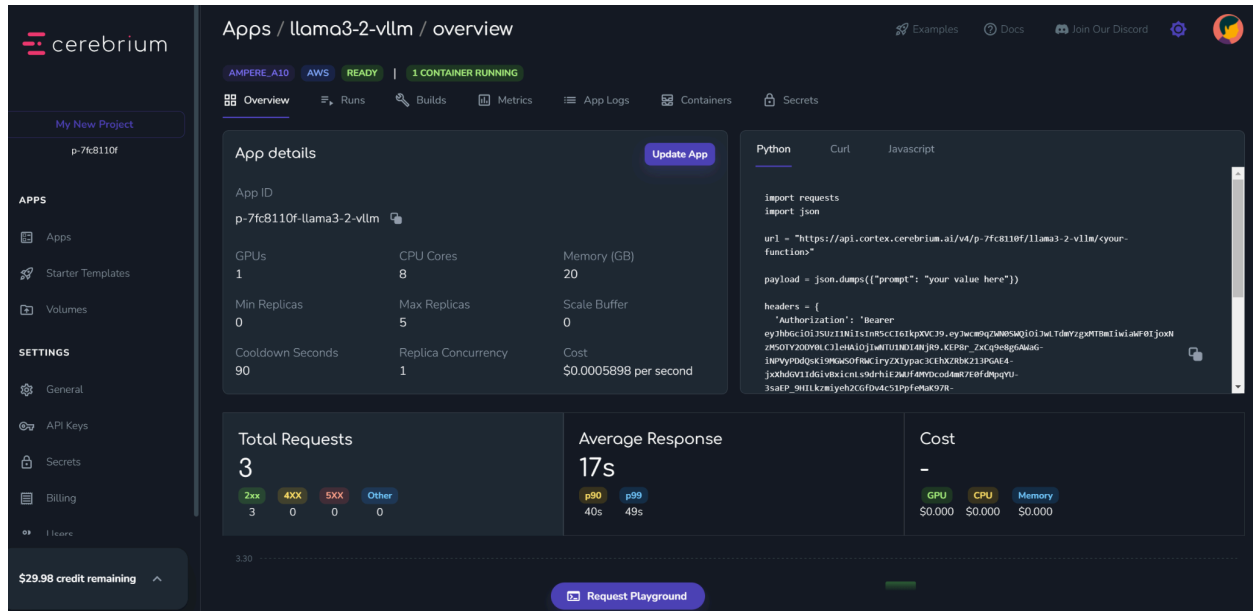
Upon successful deployment the cli will show the Endpoint



The terminal screenshot shows the following output:
18:53:37 App started successfully!
❯ Building App...
llama3-2-vllm is now live!
App Dashboard: <https://dashboard.cerebrium.ai/projects/p-7fc8110f/apps/p-7fc8110f-llama3-2-vllm>
Endpoints:
POST https://api.cortex.cerebrium.ai/v4/p-7fc8110f/llama3-2-vllm/{function_name}

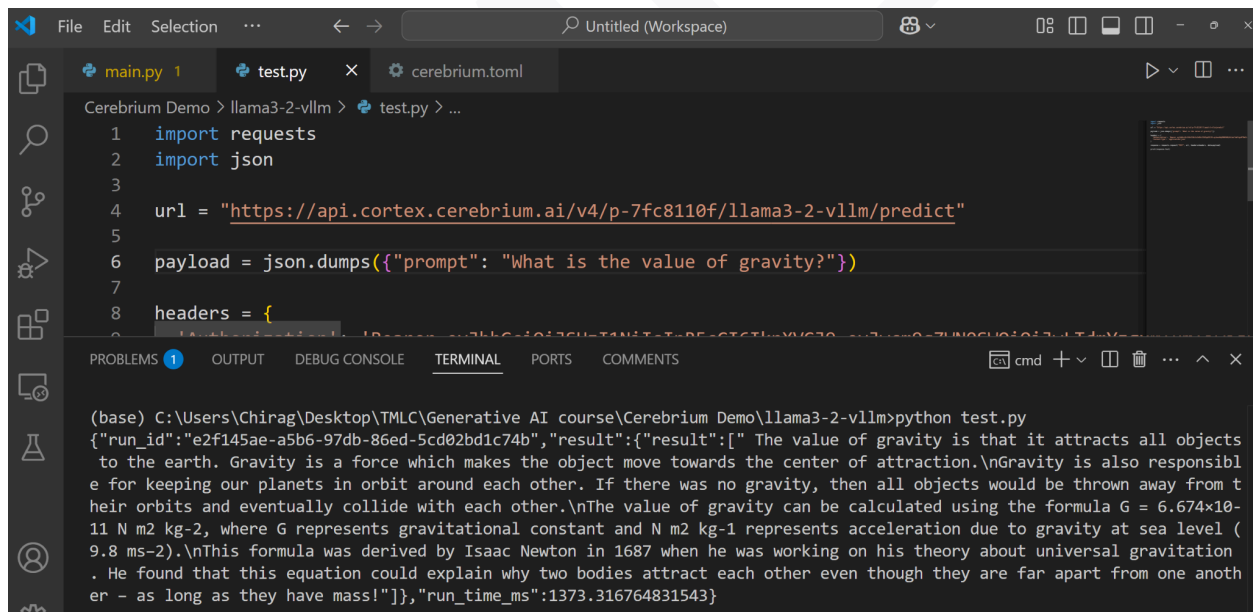
8. Inference

The overview of the app created will show the Python inference code.



The screenshot shows the Cerebrum dashboard for an application named 'llama3-2-vllm'. The dashboard is divided into several sections:

- App details:** Shows the App ID 'p-7fc8110f-llama3-2-vllm'. It lists hardware specifications: 1 GPU, 8 CPU Cores, and 20 GB Memory. It also shows configuration details: Min Replicas 0, Max Replicas 5, Scale Buffer 0, and a Cost of \$0.0005898 per second.
- Performance Metrics:** Displays 'Total Requests' as 3 (with a breakdown of 3 2xx, 0 4xx, 0 5xx, and 0 Other) and 'Average Response' time as 17s (with a breakdown of p90 at 40s and p99 at 49s).
- Cost:** Shows a total cost of \$0.000, broken down by GPU, CPU, and Memory.
- Python Inference Code:** A code editor showing the Python code for the inference endpoint. It includes imports for 'requests' and 'json', a URL for the Cerebrum API, a payload for a prompt, and headers for authentication.



The screenshot shows a code editor with a Python file named 'test.py'. The code is as follows:

```
1 import requests
2 import json
3
4 url = "https://api.cortex.cerebrum.ai/v4/p-7fc8110f/llama3-2-vllm/predict"
5
6 payload = json.dumps({"prompt": "What is the value of gravity?"})
7
8 headers = {
```

The terminal output shows the result of running the code:

```
(base) C:\Users\Chirag\Desktop\TMLC\Generative AI course\Cerebrum Demo\llama3-2-vllm>python test.py
{"run_id": "e2f145ae-a5b6-97db-86ed-5cd02bd1c74b", "result": {"result": ["The value of gravity is that it attracts all objects to the earth. Gravity is a force which makes the object move towards the center of attraction.\nGravity is also responsible for keeping our planets in orbit around each other. If there was no gravity, then all objects would be thrown away from their orbits and eventually collide with each other.\nThe value of gravity can be calculated using the formula G = 6.674x10^-11 N m^2 kg^-2, where G represents gravitational constant and N m^2 kg^-1 represents acceleration due to gravity at sea level (9.8 ms^-2).\nThis formula was derived by Isaac Newton in 1687 when he was working on his theory about universal gravitation. He found that this equation could explain why two bodies attract each other even though they are far apart from one another - as long as they have mass!"], "run_time_ms": 1373.316764831543}]}
```

The dashboard will also provide information like total requests, latency in requests and cost.

Download the code from [here](#)